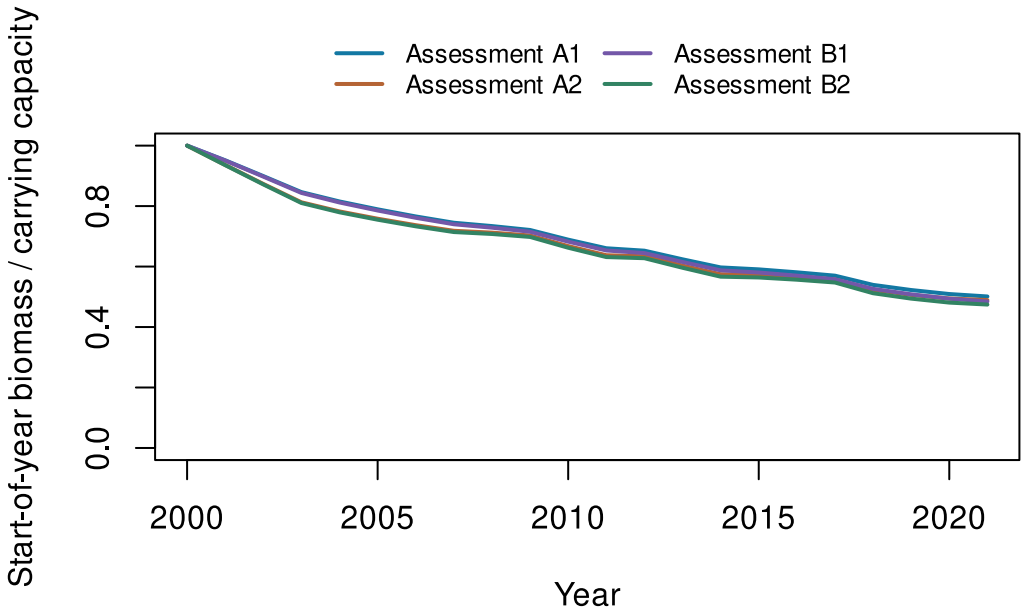
Surplus-production assessment report

Compare how two CPUE indices and two fixed growth assumptions affect the fitted biomass trajectory.

Path: Annual index and catches → fixed-growth Schaefer fit → biomass trajectory.

Biomass comparison



Each trajectory shows start-of-year biomass divided by that case's fitted carrying capacity. The trajectories describe the supplied index, catches and fixed model assumptions. Differences among cases are not an uncertainty interval or stock-management advice.

Final annual values and fit checks

Case	Year	B/K	Harvest rate
Assessment A1	2021	0.501	0.136
Assessment A2	2021	0.489	0.184
Assessment B1	2021	0.484	0.143
Assessment B2	2021	0.474	0.192

Case	Growth (r)	Balance	Feasible	Gradient	Boundary	Convergence
Assessment A1	0.25	Pass	Pass	Pass	FALSE	0
Assessment A2	0.35	Pass	Pass	Pass	FALSE	0
Assessment B1	0.25	Pass	Pass	Pass	FALSE	0
Assessment B2	0.35	Pass	Pass	Pass	FALSE	0

Convergence, biomass balance and positive-biomass feasibility check the fitted calculation. They do not establish biological adequacy or validate the CPUE as a stock-wide abundance measure.

Methods

Source coverage: Fiji fleet; six observed cell–quarter strata, 2000–2022. **Annual catch:** South Pacific albacore nominal catch estimates; all gears and fleets.

The selected Fiji fleet index is not a validated stock-wide abundance index. Using it with stock-wide catches is a teaching assumption.

The analysed series runs from 2000 to 2021. Four cases combine indices A and B with fixed intrinsic growth rates of 0.25 and 0.35 per year. Biomass is the start-of-year, pre-catch state. The annual Schaefer model starts it at carrying capacity, adds surplus production and subtracts recorded catch in tonnes. R/RTMB fits carrying capacity by minimising squared log-index residuals within feasible bounds; log catchability is the mean difference between log index and log biomass. Initial biomass and growth are assumptions, rather than additional fitted parameters.

Analysis record

Produced in **Run 002**. Retained inputs keep their original producing runs.

Execution: GitHub Actions.

Calculation image: `sha256:9dea950a713b…` (full digest in the run bundle).

Container package

Coordinator source at aecbf8e3

Assessment component source at 0e29121c

Annual values are available in the **Data** tab. Input files, settings, source versions and checksums are available in **Inputs & versions** and the downloaded run. Rendering reads saved results and does not fit models again.